UI BUILDER

AUTOMATIC CODE GENERATION TOOL TO DESIGN AND DEVELOP UI

Project ID: 2022-070



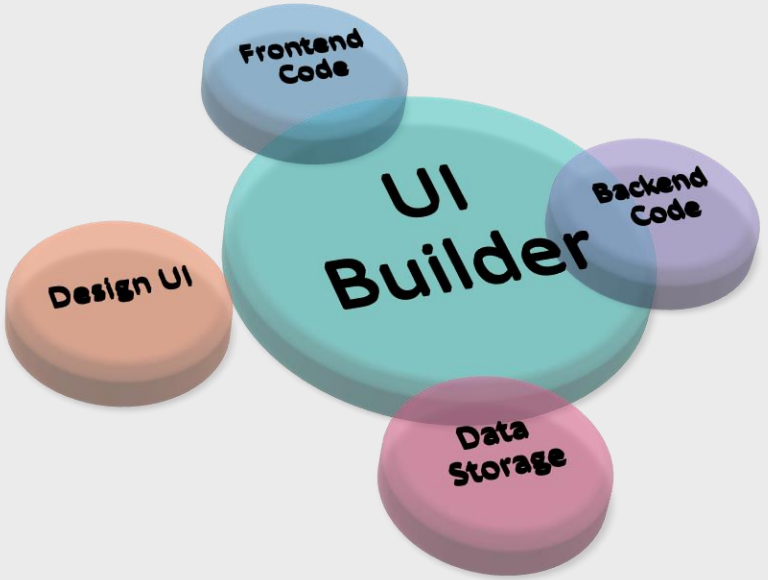
SLIIT
FACULTY OF COMPUTING

TEAM MEMBERS

Group ID: 2022-070	
Supervisor: Dr. Nuwan Kodagoda	
Co Supervisor: Ms. Kushnara Suriyawansa	
Student Name	**Student ID**
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OUR RESEARCH

- Get insights to Automatic Code generation (ACG) concepts?
- Critical analysis of existing ACG tools and their usage in industry.
- Introduce a new ACG tool with enhanced features.
- Promote the introduced tool and deploy on production environment.



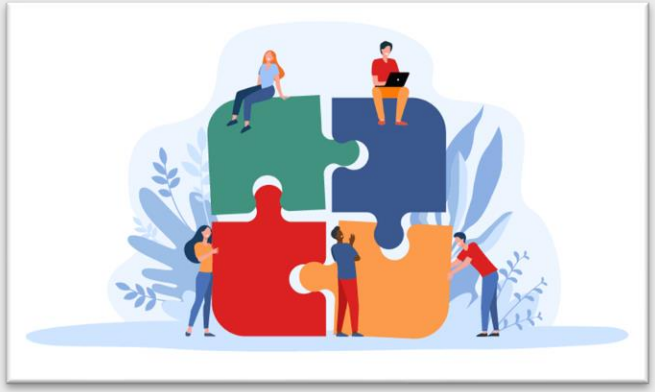
RESEARCH PROBLEM



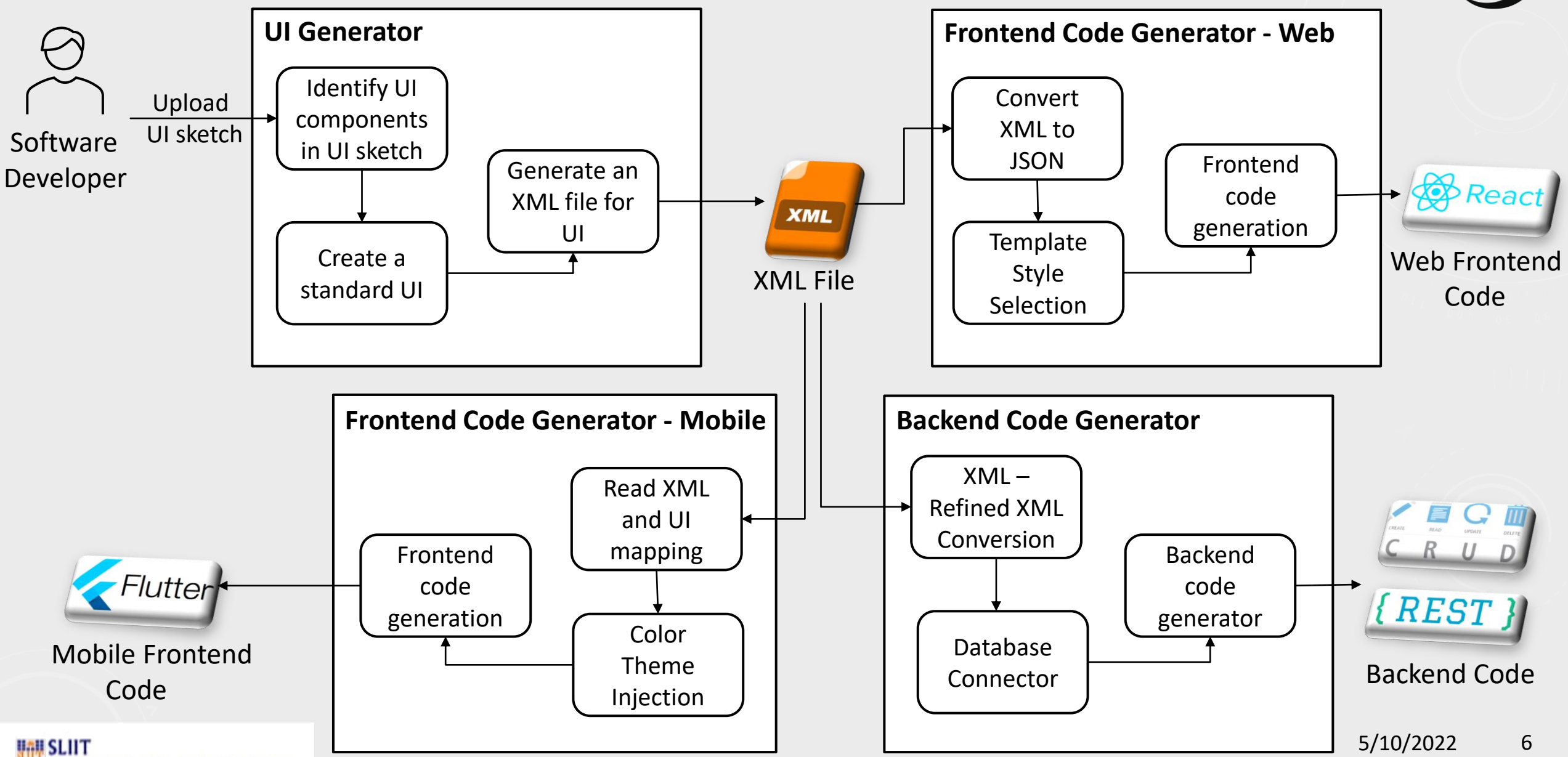
1. Need to follow the traditional method of software engineering process.
2. Available tools don’t fulfill the real requirements of software developers.
3. Auto-generated Code do not have the texture of real-handwritten code.
4. Auto-generated code only focuses on frontend, whereas backend code still needs to be written manually.

RESEARCH OBJECTIVES

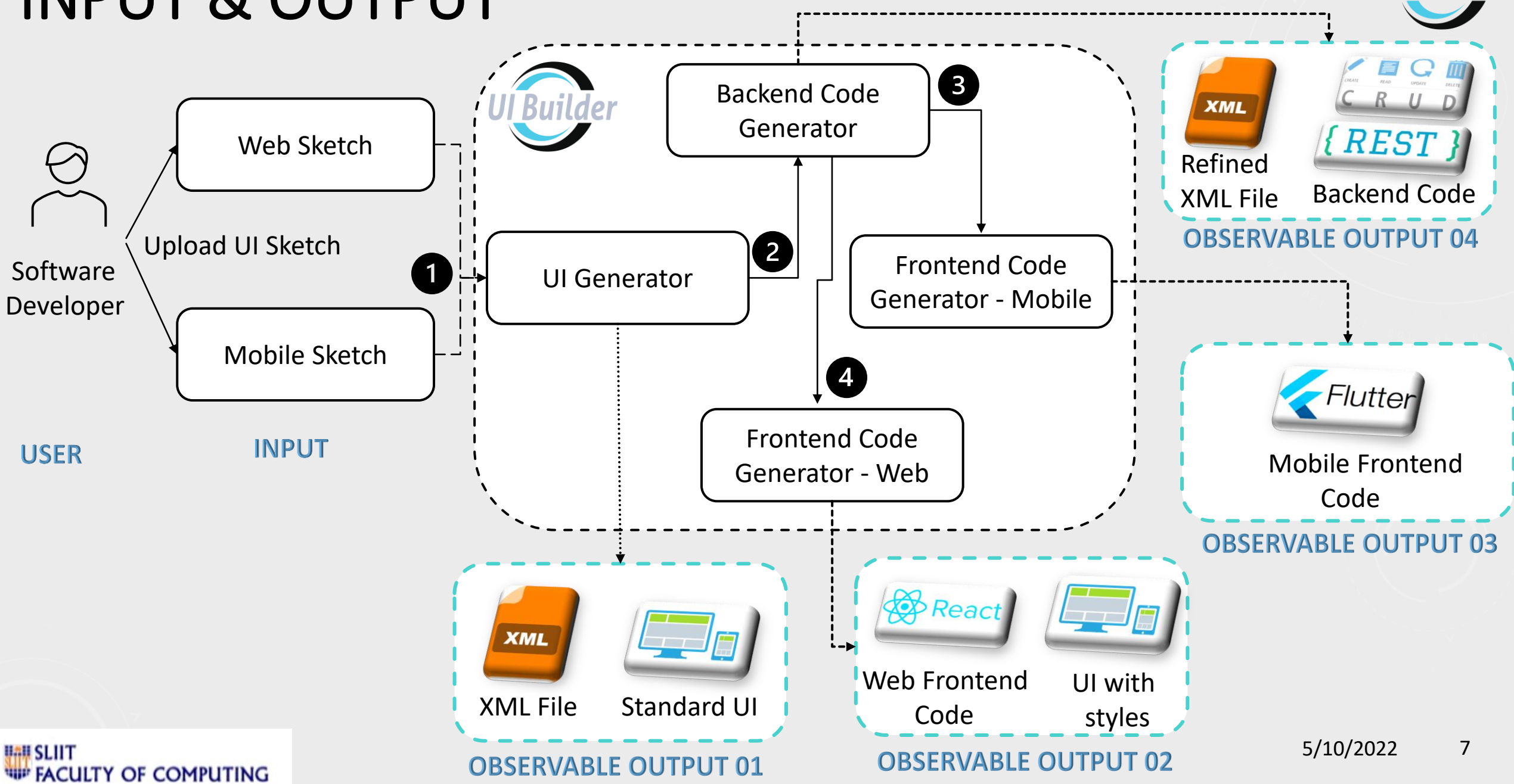
- Generate UI templates according to UI rough sketches given by users.
- Generate web frontend code from UI template using MERN Stack technologies.
- Generate mobile frontend code from UI template using code mapping.
- Develop backend code and DB generator adhering to software engineering principles.



SYSTEM OVERVIEW DIAGRAM



INPUT & OUTPUT



OUR PLAN TO PROMOTE THE TOOL



- Conducting a live demonstration of code generation process in developer environment.
- Understanding the need and expectations of developers through automated feedback services integrated with the tool.
- Select startup companies and describe them the capabilities and advantages of using UI builder tool for their software development process.
- Use social media and publish an official website for the tool to make it reach out for wide range of people.
- Conduct a workshop on the quick setup of the tool in their PC for 1st and 2nd year IT students.
- Speak around '**Anyone Can Code**' concept to make It stand unique among the other code generation tools.
- Use quick startup guides and YouTube tutorials for better usability and user-friendliness.



IT19150758|EKANAYAKE K.L.W

BSC (HONS) DEGREE IN INFORMATION TECHNOLOGY

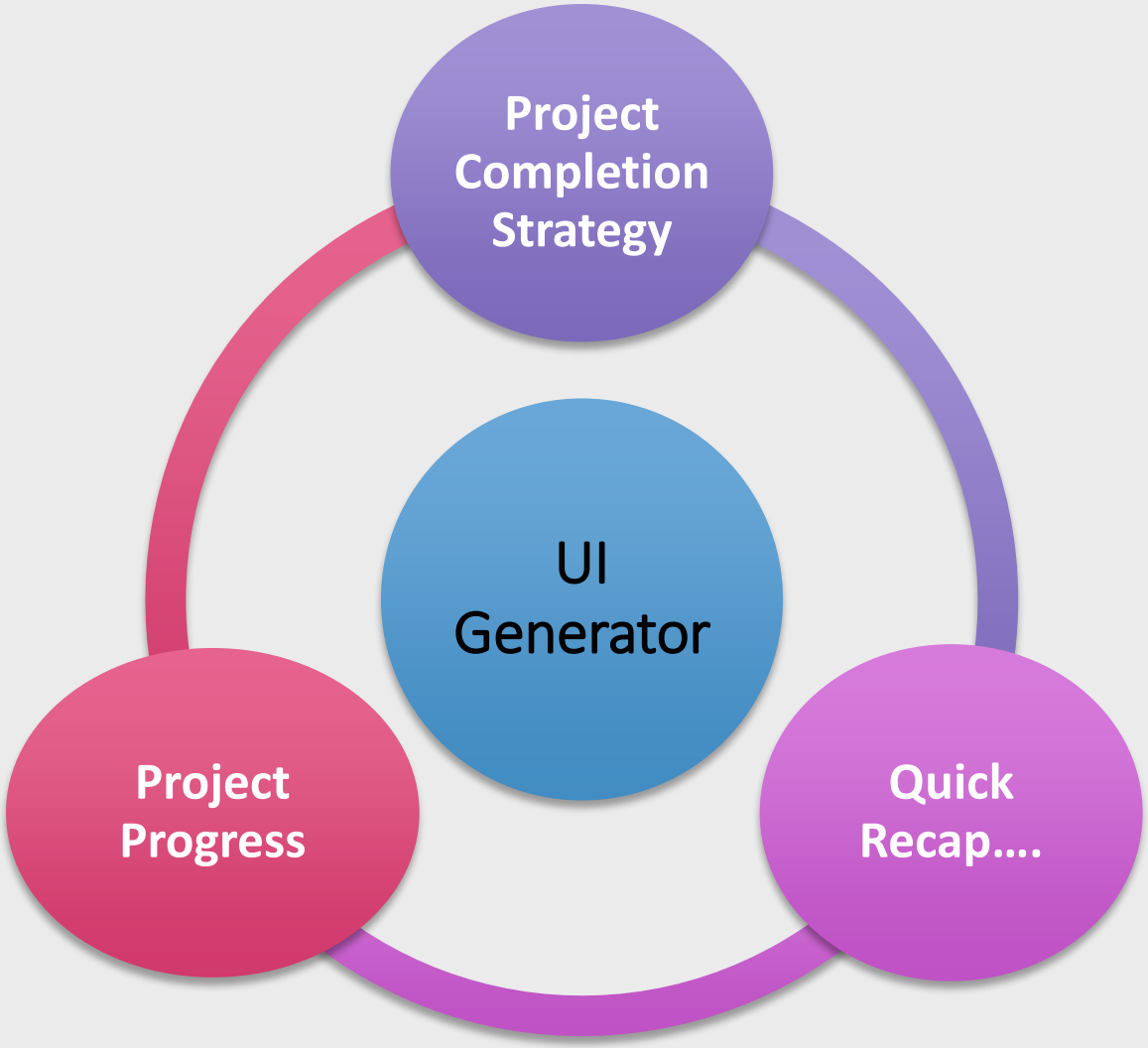
(SPECIALIZATION IN SOFTWARE ENGINEERING)

RESEARCH COMPONENT

GENERATING STANDARD UIs and XML files ACCORDING TO ROUGH SKETCHES.



INTRODUCTION



QUICK RECAP.....

Research



- Introduce a standard UI template generator according to the rough Uis.

Research Question



- How to generate standard UI according to rough UIs.

Objectives



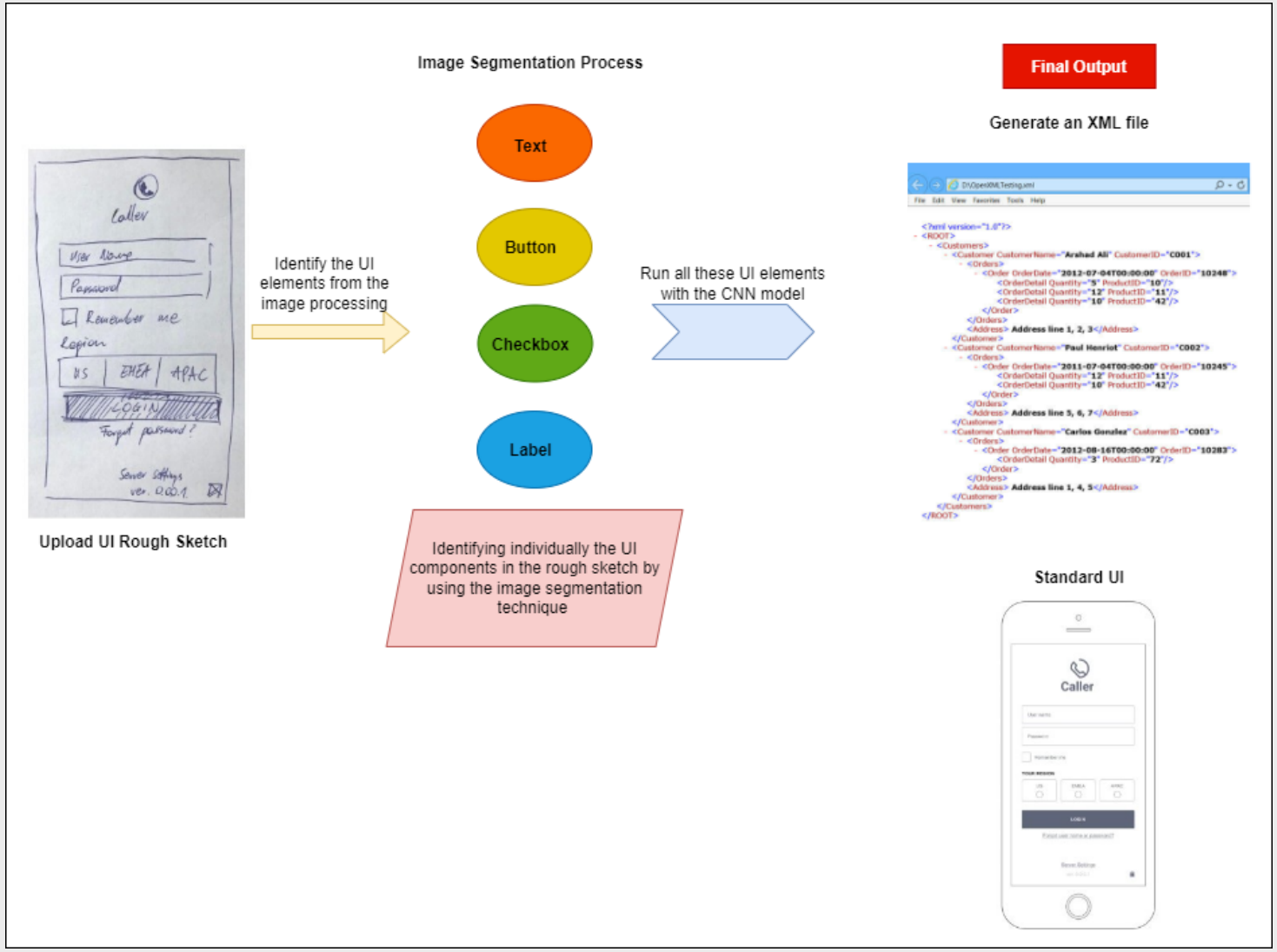
- Generate a standard UIs and XML files according to rough UIs.

Research Gap

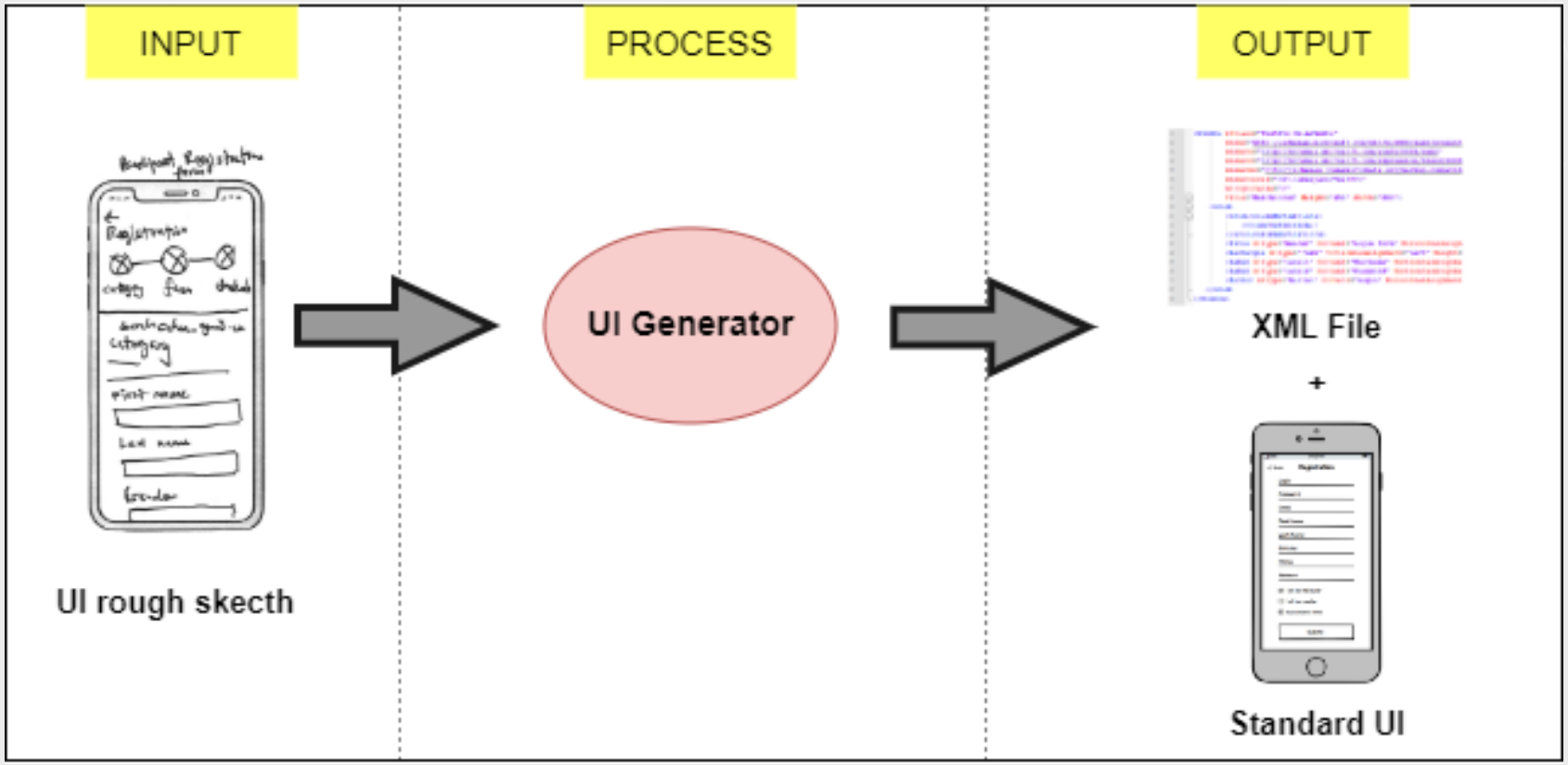


- Existing UI generator tools unable to convert rough UIs to standard Uis.

PROCESS


Image Segmentation Process
Final Output
Generate an XML file
Text
Button
Checkbox
Label
Identify the UI elements from the image processing
Run all these UI elements with the CNN model
Upload UI Rough Sketch
Identifying individually the UI components in the rough sketch by using the image segmentation technique
Standard UI
Caller

INPUT OUTPUT



PROGRESS COMPLETION

Collected the rough UI dataset and tested , trained.

- Collected the dataset from a survey.
- Trained and tested using a CNN model

UI elements predictor.

- Upload the UI sketches to application.
- Hit the rough sketches to the model and get the predicted results.

User input form module.

- Getting the predicted values for sketches and generate a dynimically changeable user input form.
- User can give elements values.

Generate standard UI module.

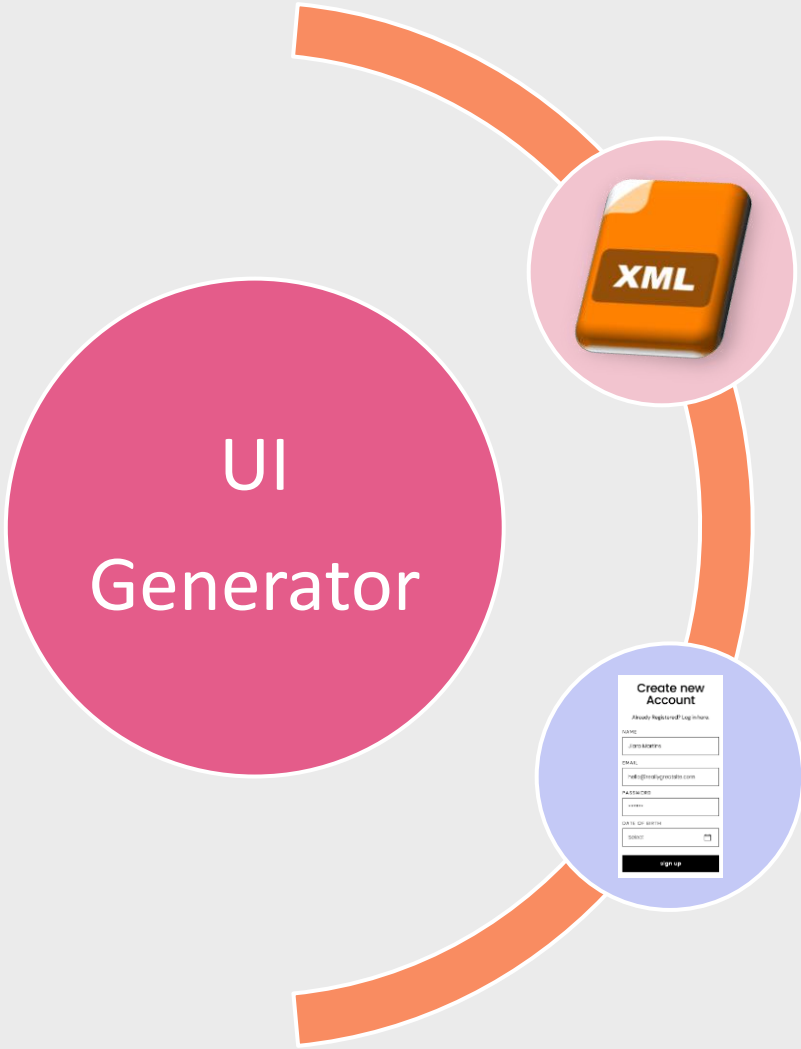
- Using the CNN model -> get the predicted results and generate standard UIs.

Generate XML file module and download XML file process.

- Generate the XML files according to the standard UI elements.
- From the generated XML file download function will execute.



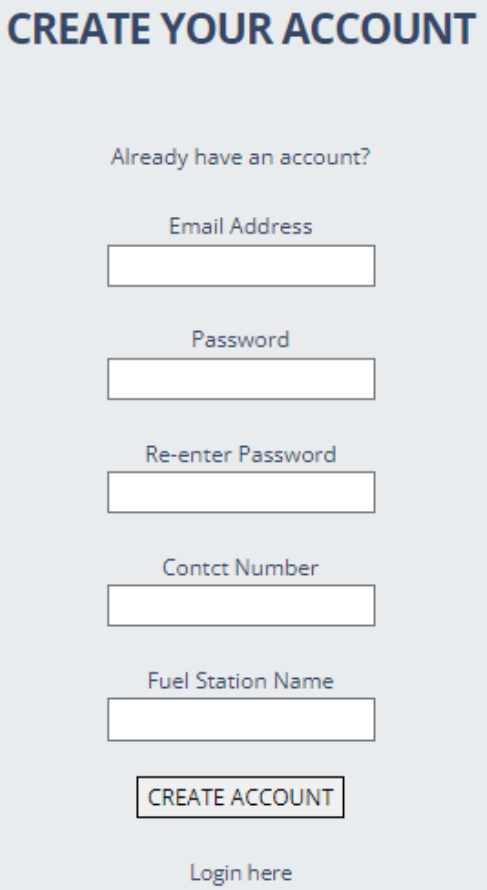
OBSERVABLE OUTCOMES



Generated XML File

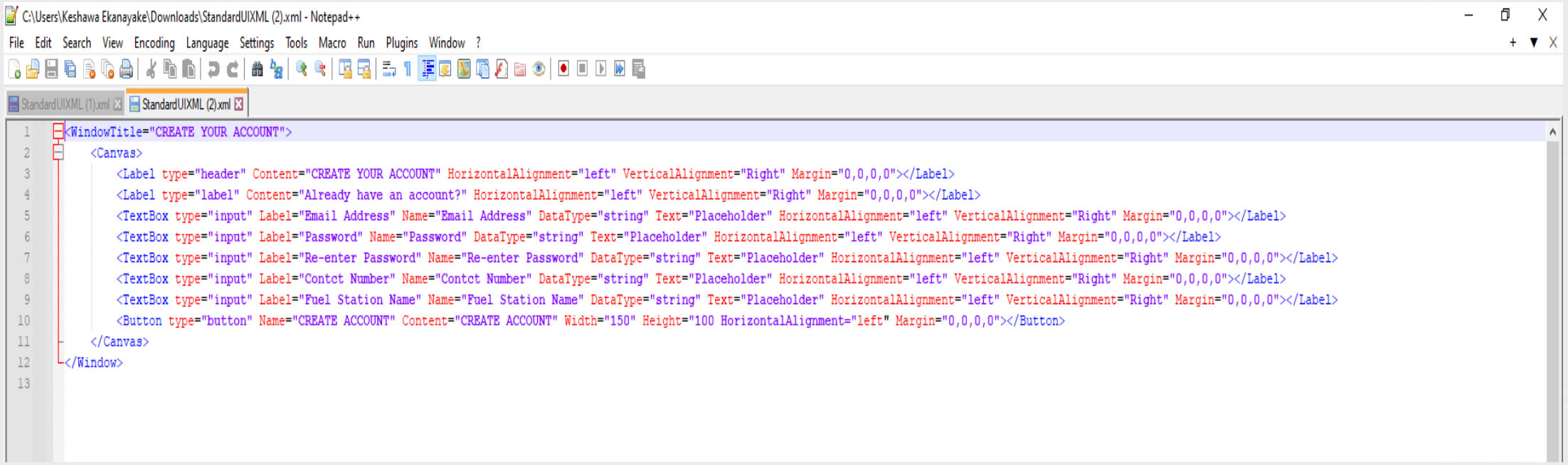
Standard UI

SAMPLE GENERATED XML CODE & STANDARD UI - VIEW IN VS CODE




```
<WindowTitle="CREATE YOUR ACCOUNT">
    <Canvas>
        <Label type="header" Content="CREATE YOUR ACCOUNT" HorizontalAlignment="left"
VerticalAlignment="Right" Margin="0,0,0,0"></Label>
        <Label type="label" Content="Already have an account?" HorizontalAlignment="left"
VerticalAlignment="Right" Margin="0,0,0,0"></Label>
        <TextBox type="input" Label="Email Address" Name="Email Address" DataType="string"
Text="Placeholder" HorizontalAlignment="left" VerticalAlignment="Right" Margin="0,0,0,0"></Label>
        <TextBox type="input" Label="Password" Name="Password" DataType="string" Text="Placeholder"
HorizontalAlignment="left" VerticalAlignment="Right" Margin="0,0,0,0"></Label>
        <TextBox type="input" Label="Re-enter Password" Name="Re-enter Password" DataType="string"
Text="Placeholder" HorizontalAlignment="left" VerticalAlignment="Right" Margin="0,0,0,0"></Label>
        <TextBox type="input" Label="Contct Number" Name="Contct Number" DataType="string"
Text="Placeholder" HorizontalAlignment="left" VerticalAlignment="Right" Margin="0,0,0,0"></Label>
        <TextBox type="input" Label="Fuel Station Name" Name="Fuel Station Name" DataType="string"
Text="Placeholder" HorizontalAlignment="left" VerticalAlignment="Right" Margin="0,0,0,0"></Label>
        <Button type="button" Name="CREATE ACCOUNT" Content="CREATE ACCOUNT" Width="150"
Height="100 HorizontalAlignment="left" Margin="0,0,0,0"></Button>
    </Canvas>
</Window>
```

SAMPLE GENERATED XML CODE - VIEW IN APPLICATION.





IT19158228|SAKTHIVEL.S

BSC (HONS) DEGREE IN INFORMATION TECHNOLOGY

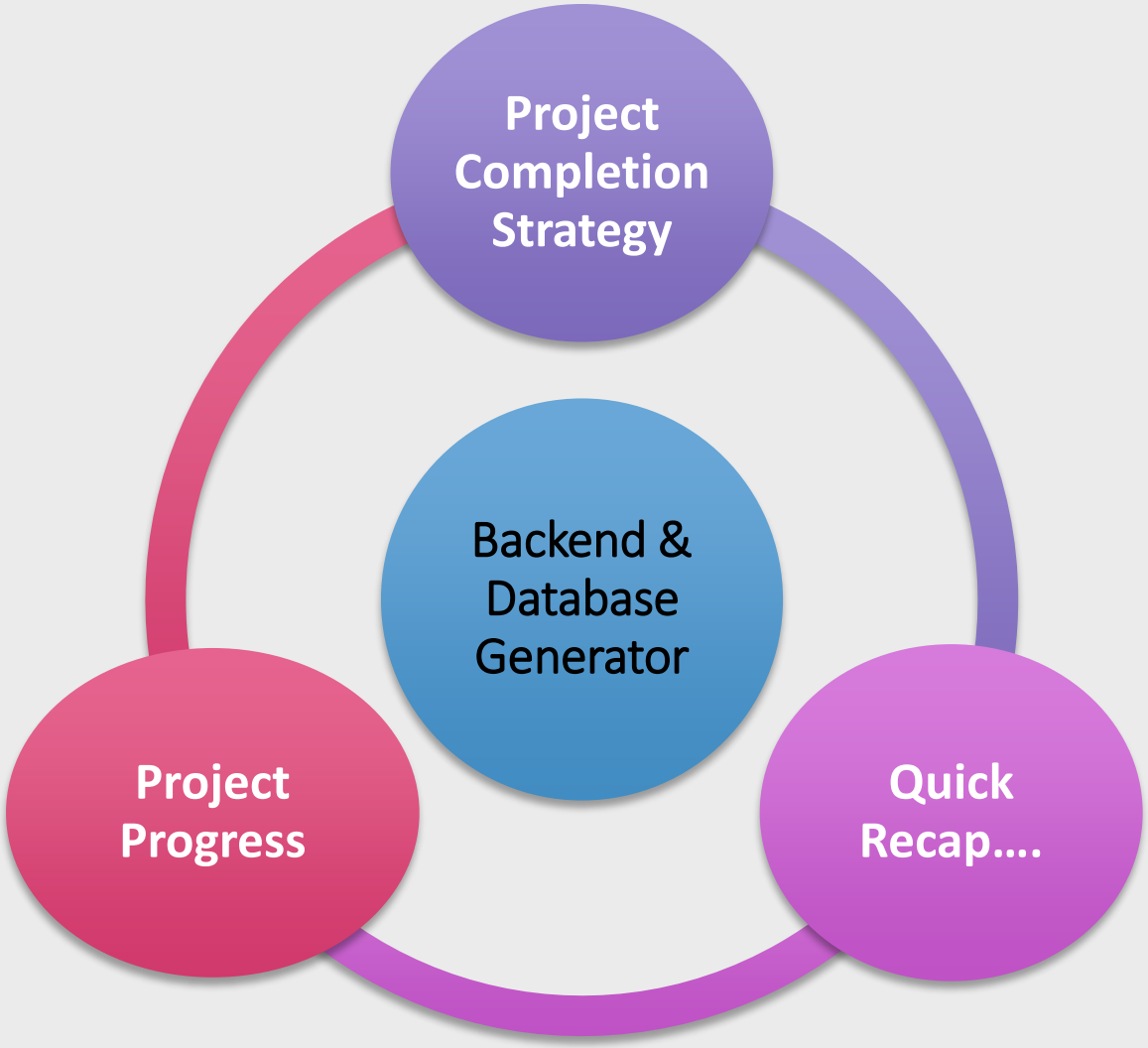
(SPECIALIZATION IN SOFTWARE ENGINEERING)

RESEARCH COMPONENT

BACKEND CODE AND DATABASE GENERATOR



INTRODUCTION



QUICK RECAP.....

Research



- Introduce a backend code generator module for ACG tool.

Research Question



- How to generate backend code for the given UI.

Objectives



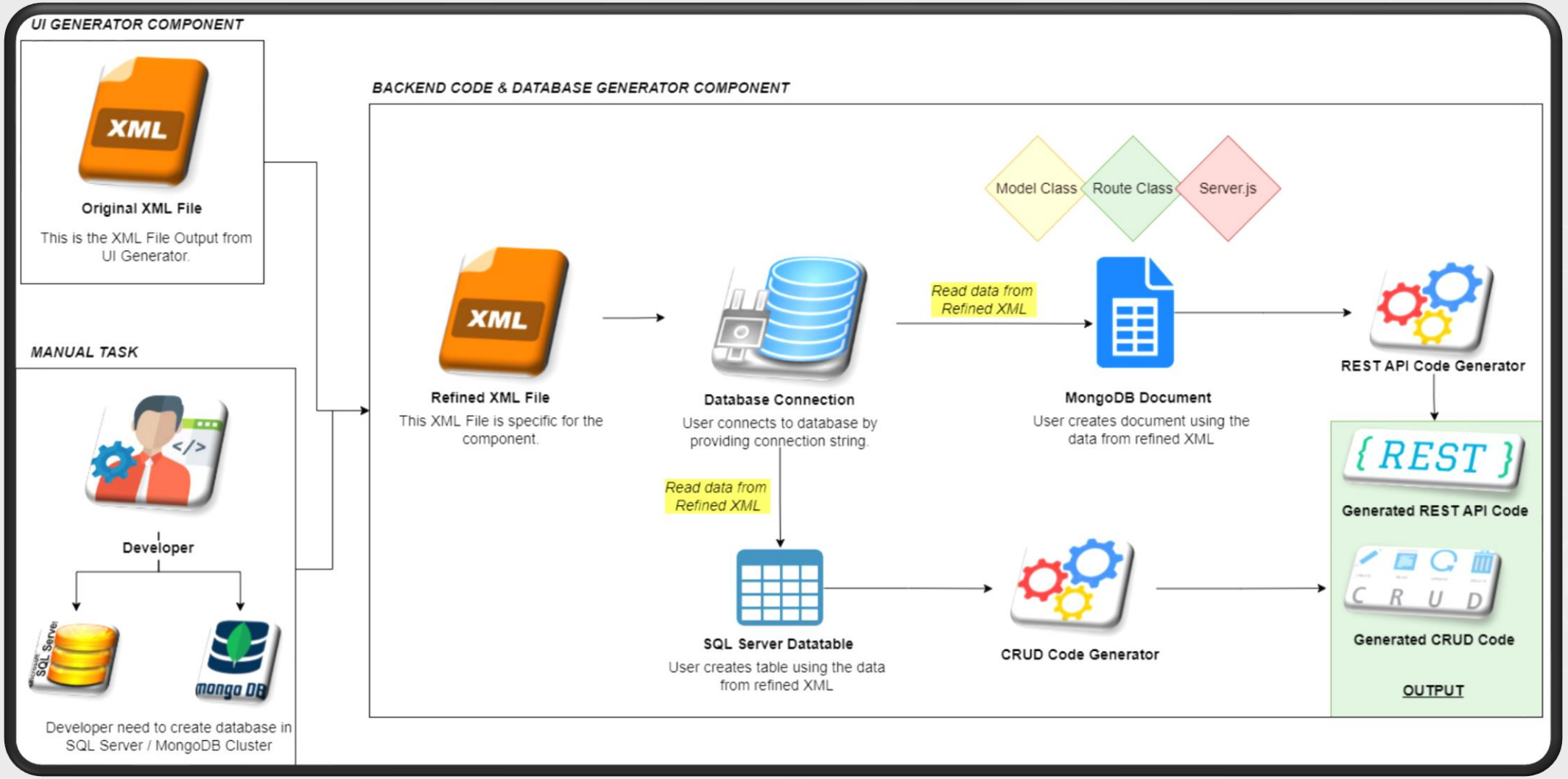
- Generate a fully functional backend code with database connectivity.

Research Gap

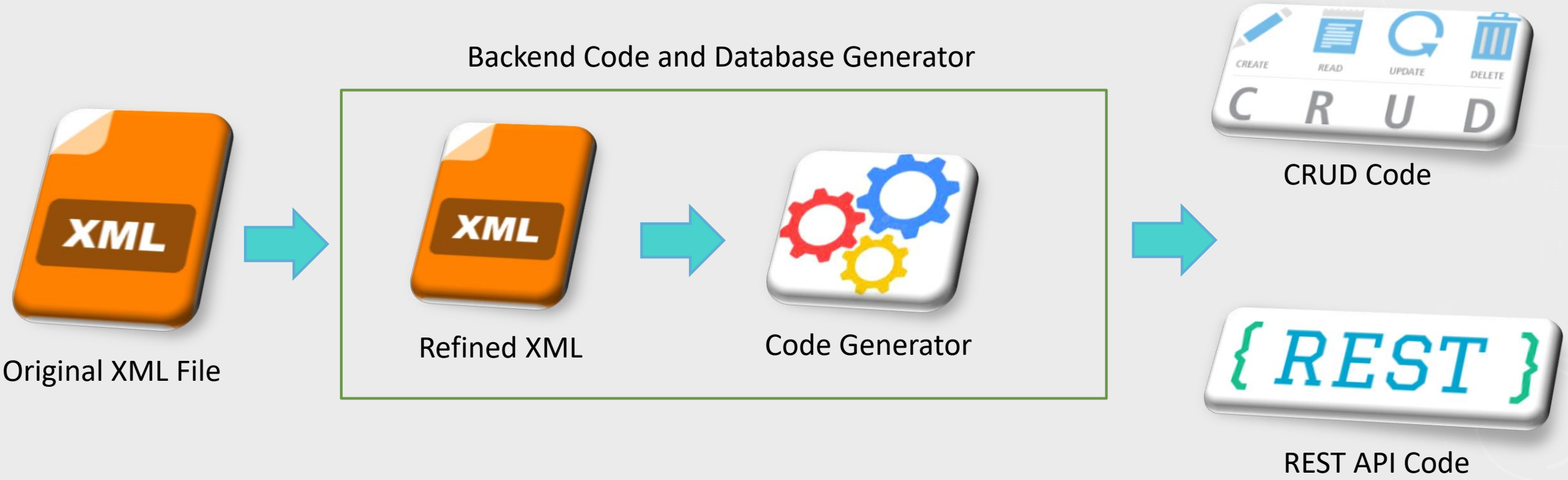


- Existing ACG tools lack backend code generation process.

PROCESS


UI GENERATOR COMPONENT
XML
Original XML File
This is the XML File Output from UI Generator.
MANUAL TASK
Developer
Developer need to create database in SQL Server / MongoDB Cluster
BACKEND CODE & DATABASE GENERATOR COMPONENT
Model Class
Route Class
Server.js
XML
Refined XML File
This XML File is specific for the component.
Database Connection
User connects to database by providing connection string.
Read data from Refined XML
MongoDB Document
User creates document using the data from refined XML
REST API Code Generator
{ REST }
Generated REST API Code
Read data from Refined XML
SQL Server Datatable
User creates table using the data from refined XML
CRUD Code Generator
C R U D
Generated CRUD Code
OUTPUT

INPUT OUTPUT



PROGRESS COMPLETION

XML to Refined XML Convertor.

- Read & process original XML file.
- Creation of Refined XML file.

Database Connector.

- Generate connection URL.
- Establish connection to Database.

Table Creation Module.

- Read created Refined XML file.
- Add or modify table columns or keys.
- Create table on Database.

CRUD Code Generator.

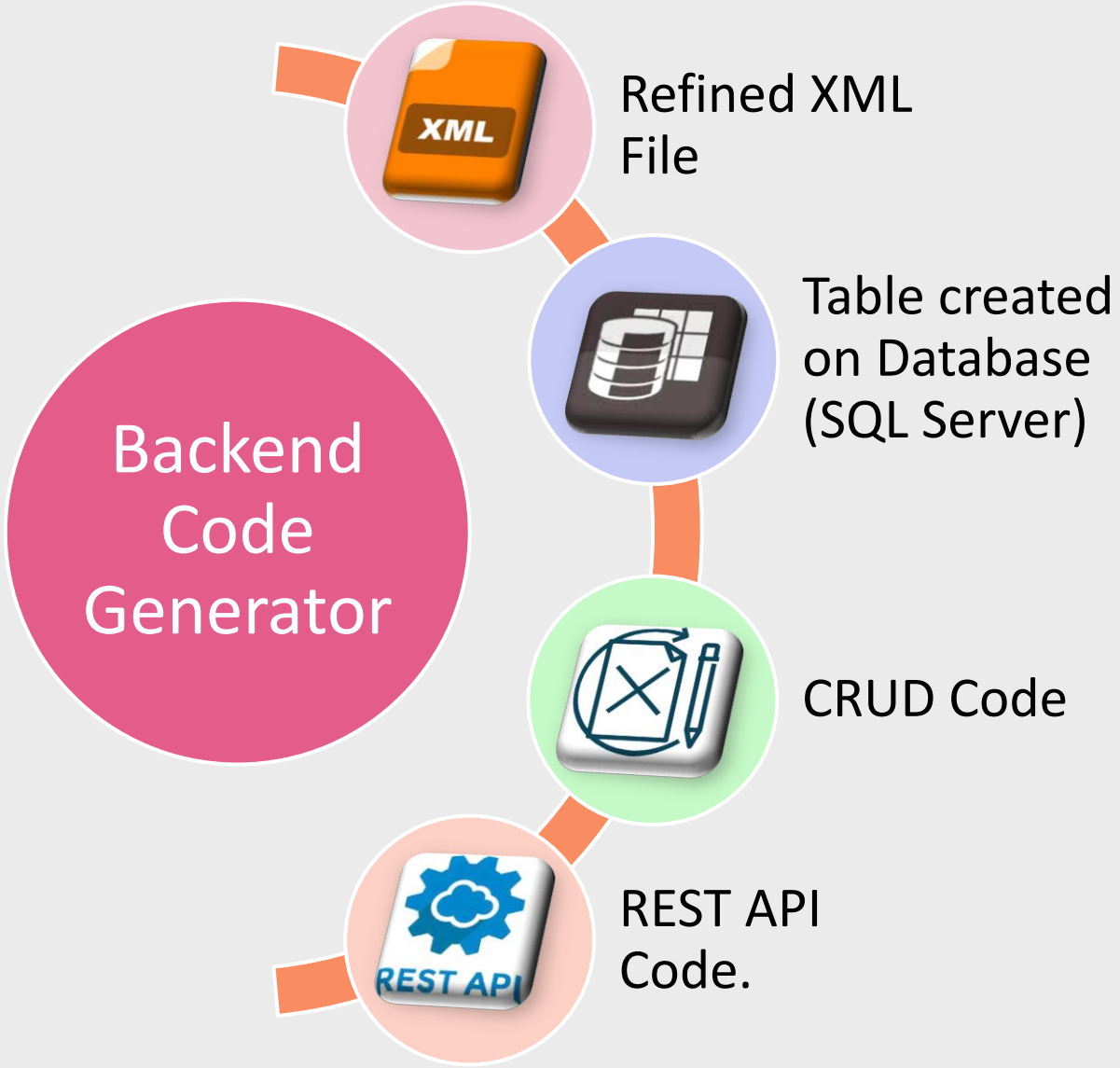
- Generate CRUD code for selected CRUD operations.
- Generate comments and embed with the code.

REST API Code Generator.

- Generate REST API code for selected HTTP methods.
- Generate comments and embed with the code.



OBSERVABLE OUTCOMES



SAMPLE GENERATED CODE - VIEW IN VS CODE

```
/*
**********************************************************************
*NOTE          :   This is an auto-generated Code
*@Developer    :   (Add Developer Name Here)
*@Purpose      :   Add Customer to database table.
*@Created Date :   5/8/2022 6:48:57 PM
**********************************************************************
*/
public void addCustomer(String custId, String custName, String custAddress, String cutsTel, String dateOfBirth) {
    try{
        SqlConnection con = new SqlConnection(@Data Source=DESKTOP-I53H1O7;Initial Catalog='Test Database';Integrated Security=True);
        SqlCommand cmd;

        String sql = "INSERT INTO Customer VALUES(@custId,@custName,@custAddress,@cutsTel,@dateOfBirth);"
        con.Open();
        cmd = new SqlCommand(sql, con);

        cmd.Parameters.AddWithValue("@custId", custId);
        cmd.Parameters.AddWithValue("@custName", custName);
        cmd.Parameters.AddWithValue("@custAddress", custAddress);
        cmd.Parameters.AddWithValue("@cutsTel", cutsTel);
        cmd.Parameters.AddWithValue("@dateOfBirth", dateOfBirth);

        cmd.ExecuteNonQuery();

        Console.Write('Date Added Successfully!');

    } catch(Exception ex) {
        Console.Write("Exception Occurred!");
    }

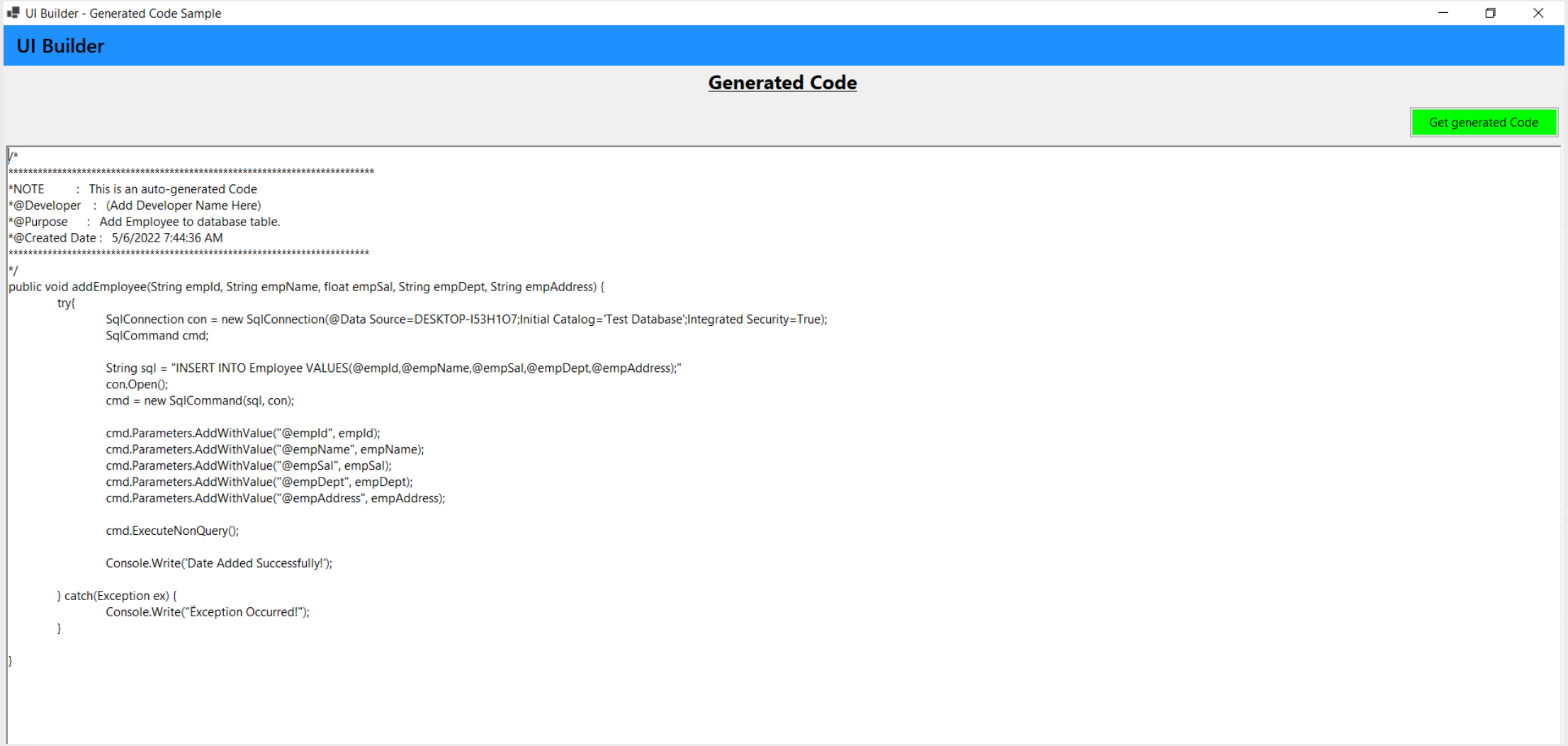
}
```



Observe!

1. **Comments.**
2. **Method Signature.**
3. **Syntax.**
4. **Arrange of Code.**
5. **Error Handling.**

SAMPLE GENERATED CODE – IN APPLICATION




```
/*
**********************************************************************
*NOTE        :  This is an auto-generated Code
*@Developer   :  (Add Developer Name Here)
*@Purpose     :  Add Employee to database table.
*@Created Date :  5/6/2022 7:44:36 AM
**********************************************************************
*/
public void addEmployee(String empId, String empName, float empSal, String empDept, String empAddress) {
        try{
                SqlConnection con = new SqlConnection(@Data Source=DESKTOP-I53H1O7;Initial Catalog='Test Database';Integrated Security=True);
                SqlCommand cmd;

                String sql = "INSERT INTO Employee VALUES(@empId,@empName,@empSal,@empDept,@empAddress);"
                con.Open();
                cmd = new SqlCommand(sql, con);

                cmd.Parameters.AddWithValue("@empId", empId);
                cmd.Parameters.AddWithValue("@empName", empName);
                cmd.Parameters.AddWithValue("@empSal", empSal);
                cmd.Parameters.AddWithValue("@empDept", empDept);
                cmd.Parameters.AddWithValue("@empAddress", empAddress);

                cmd.ExecuteNonQuery();

                Console.Write('Date Added Successfully!');

        } catch(Exception ex) {
                Console.Write("Exception Occurred!");
        }

}
```



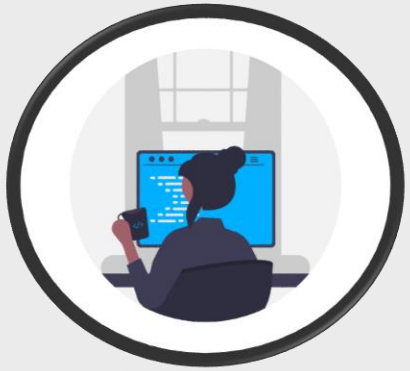
IT19144986|H.M. KASUNI NAVODYA

BSC (HONS) DEGREE IN INFORMATION TECHNOLOGY

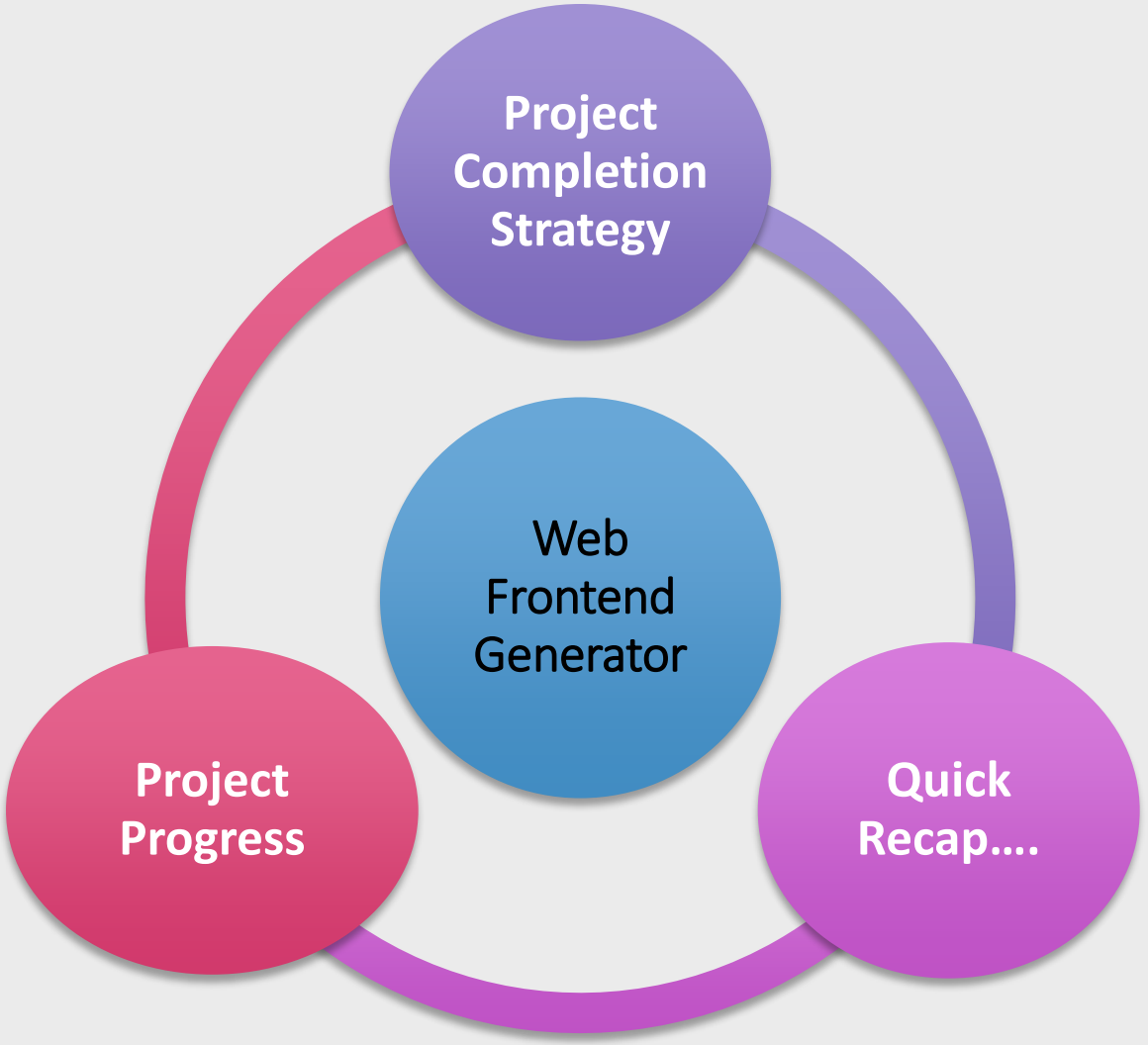
(SPECIALIZATION IN SOFTWARE ENGINEERING)

RESEARCH COMPONENT

FRONT-END CODE GENERATION FOR WEB APPLICATIONS.



INTRODUCTION



QUICK RECAP.....

Research



- Introduce a web frontend generator module for ACG tool.

Research Question



- How to generate web frontend code for the given UI.

Objectives



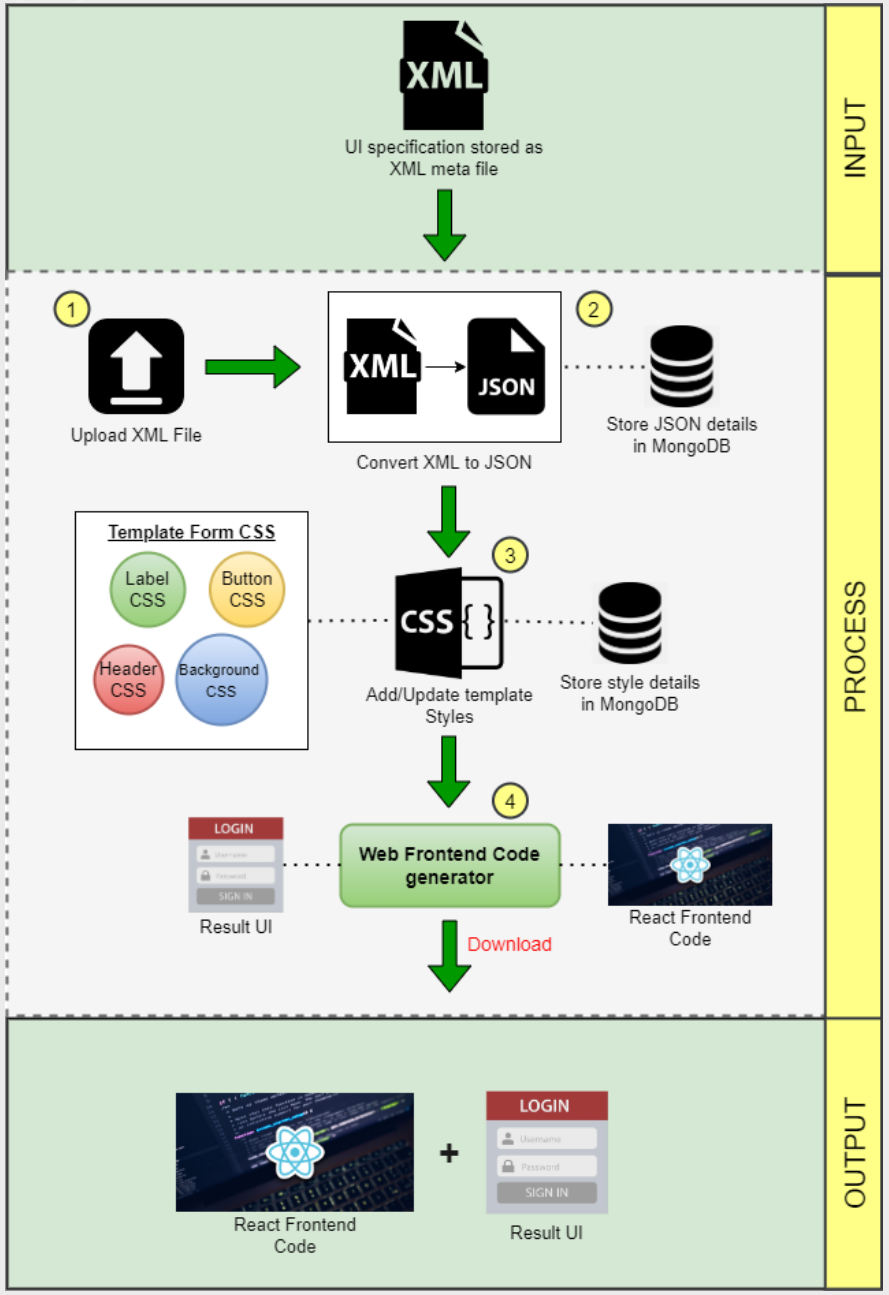
- Generate a simple and accurate automated web frontend code.

Research Gap

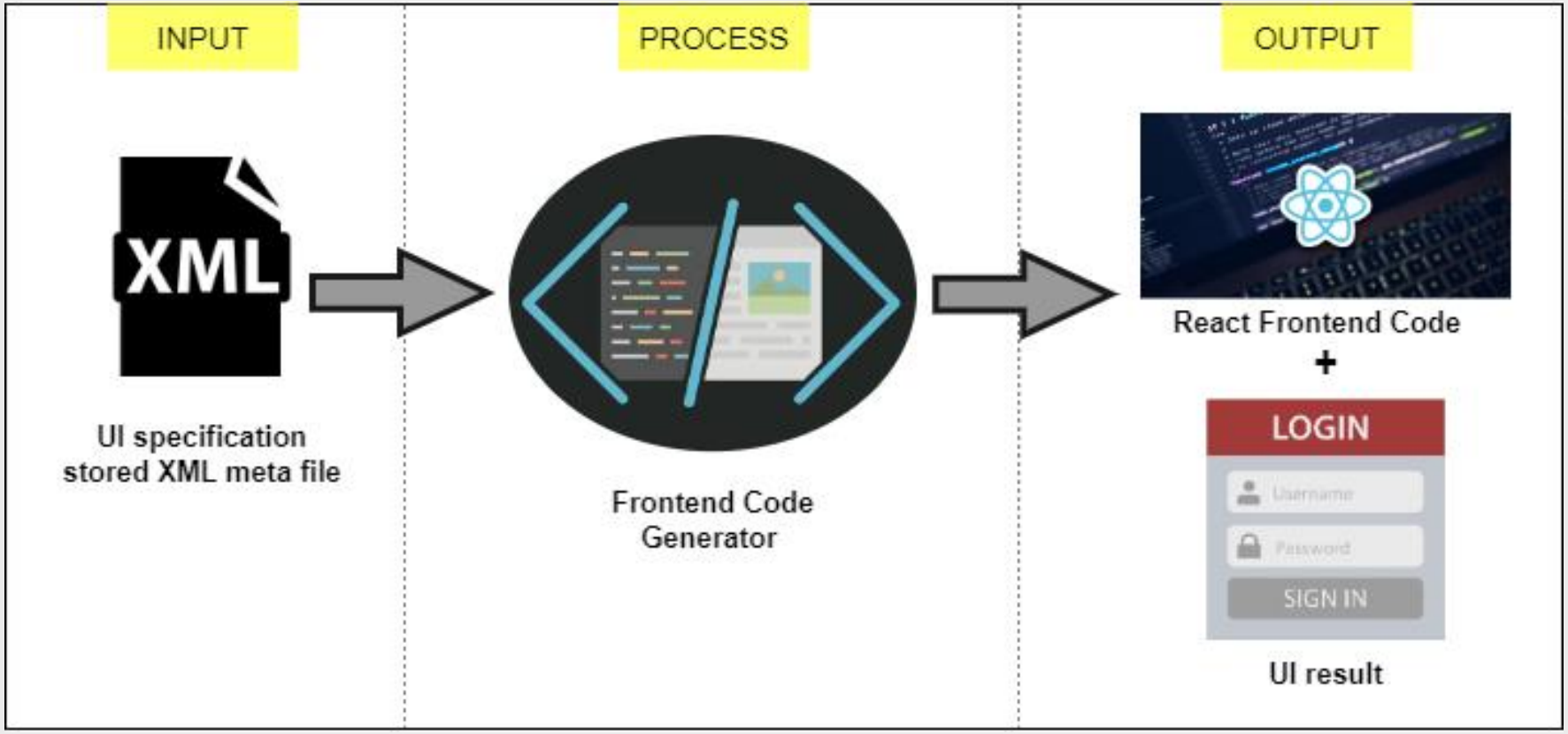


- Absence of flexible & scalable web frontend code generation process.

PROCESS


INPUT
XML
UI specification stored as XML meta file
PROCESS
1
Upload XML File
XML
JSON
Convert XML to JSON
2
Store JSON details in MongoDB
Template Form CSS
Label CSS
Button CSS
Header CSS
Background CSS
3
CSS
Add/Update template Styles
Store style details in MongoDB
4
LOGIN
Result UI
Web Frontend Code generator
React Frontend Code
Download
OUTPUT
React Frontend Code
+
LOGIN
SIGN IN
Result UI

INPUT OUTPUT



PROGRESS COMPLETION

XML to JSON Convertor

- Browse/ Drag and Drop XML File.
- Convert XML to JSON format and submit.

Template Style Selection

- Create pre-define templates according to user preferences.
- Add or modify template style details.

Web Frontend Code Generator

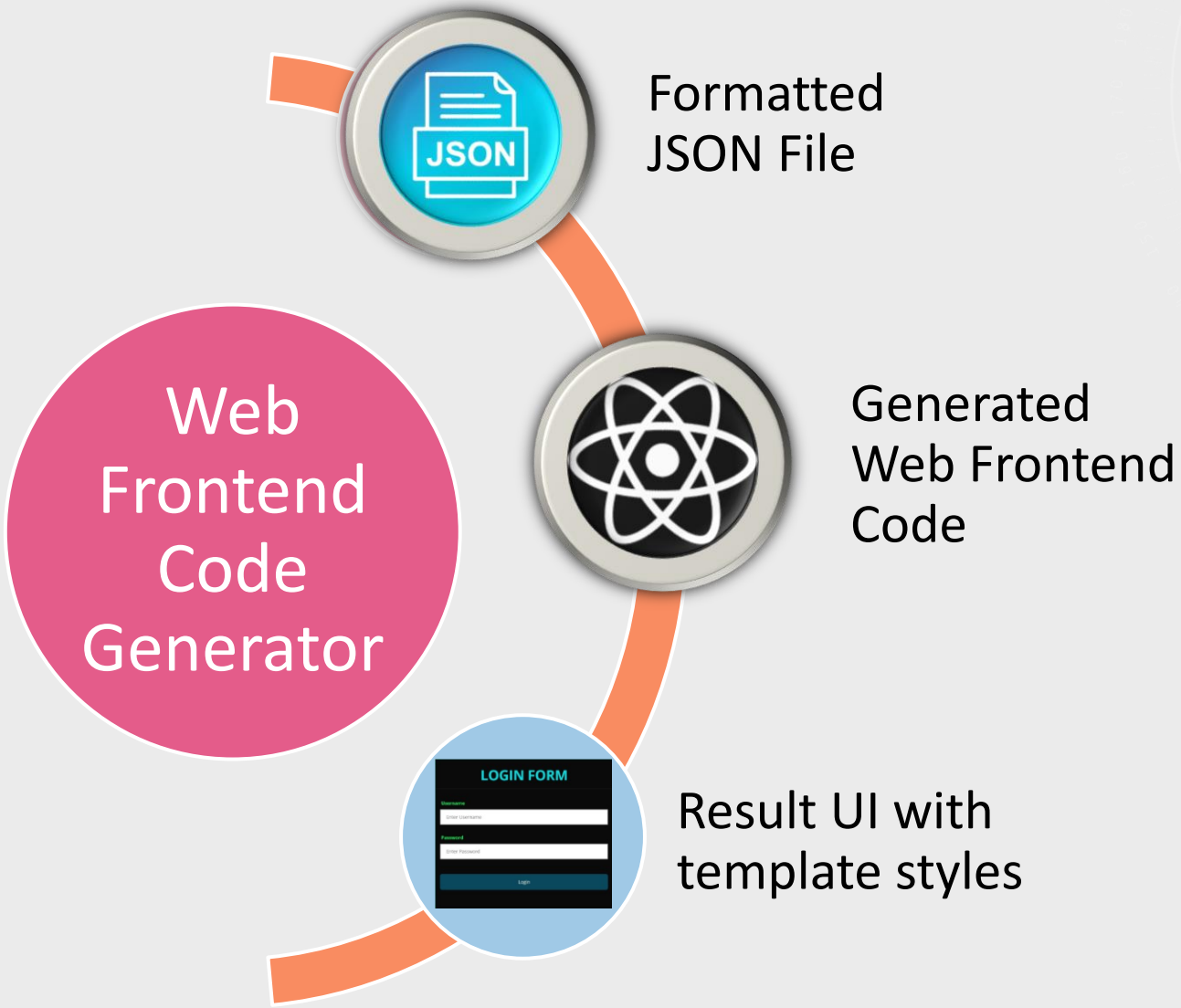
- Generate frontend code using JSON and CSS details.
- Display generated frontend code.
- Display result UI.

Download generated code and UI Result.

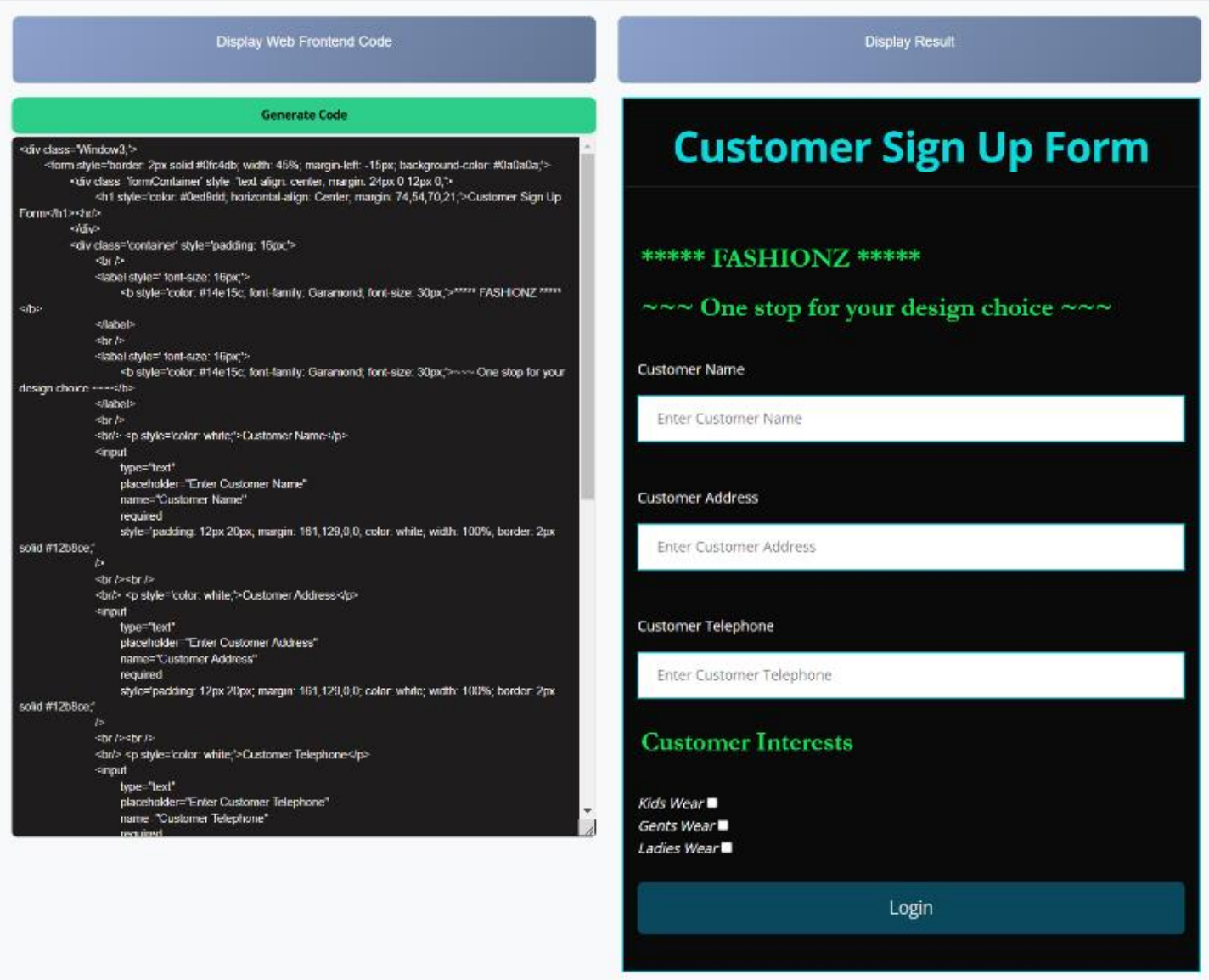
- Display generated result UI.
- Generate button for download the code(.java) and result UI(.jpg)



OBSERVABLE OUTCOMES



SAMPLE GENERATED CODE – IN APPLICATION

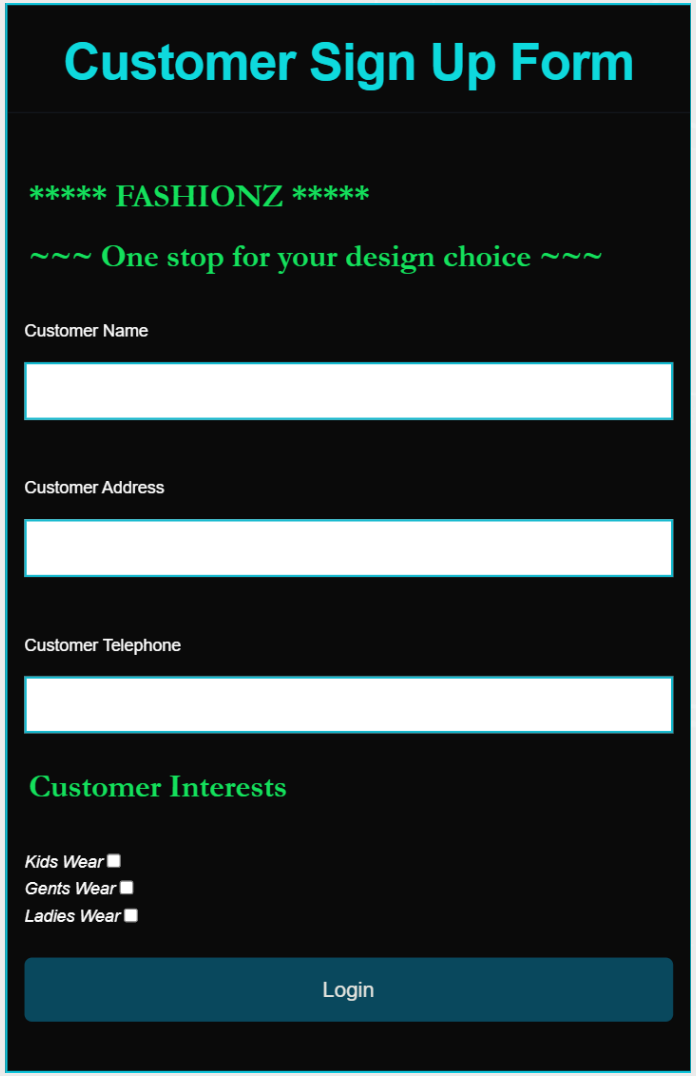


SAMPLE GENERATED CODE - VIEW IN VS CODE

Web_Frontend_Code (6).js

C: > Users > kasun > Downloads > Web_Frontend_Code (6).js

```
<div class='Window3;'>
    <form style='border: 2px solid #0fc4db; width: 45%; margin-left: -15px; background-color: #0a0a0a;'>
        <div class='formContainer' style='text-align: center; margin: 24px 0 12px 0;'>
            <h1 style='color: #0ed9dd; horizontal-align: Center; margin: 74,54,70,21;'>Customer Sign Up Form</h1><hr/>
        </div>
        <div class='container' style='padding: 16px;'>
            <br />
            <label style=' font-size: 16px;'>
                <b style='color: #14e15c; font-family: Garamond; font-size: 30px;'>***** FASHIONZ *****</b>
            </label>
            <br />
            <label style=' font-size: 16px;'>
                <b style='color: #14e15c; font-family: Garamond; font-size: 30px;'>~~~ One stop for your design choice ~~~</b>
            </label>
            <br />
            <br/> <p style='color: white;'>Customer Name</p>
            <input
                type="text"
                placeholder="Enter Customer Name"
                name="Customer Name"
                required
                style='padding: 12px 20px; margin: 161,129,0,0; color: white; width: 100%; border: 2px solid #12b8ce;'
            />
            <br /><br />
            <br/> <p style='color: white;'>Customer Address</p>
            <input
                type="text"
                placeholder="Enter Customer Address"
                name="Customer Address"
                required
                style='padding: 12px 20px; margin: 161,129,0,0; color: white; width: 100%; border: 2px solid #12b8ce;'
            />
            <br /><br />
            <br/> <p style='color: white;'>Customer Telephone</p>
```





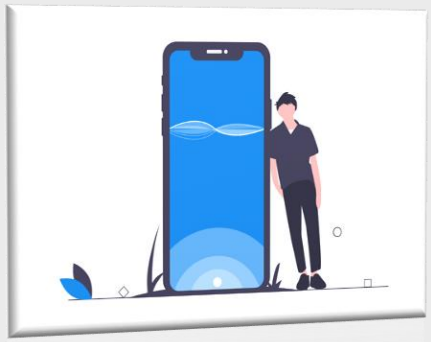
IT19150826|D.P. KAVINDI GIMSHANI

BSC (HONS) DEGREE IN INFORMATION TECHNOLOGY

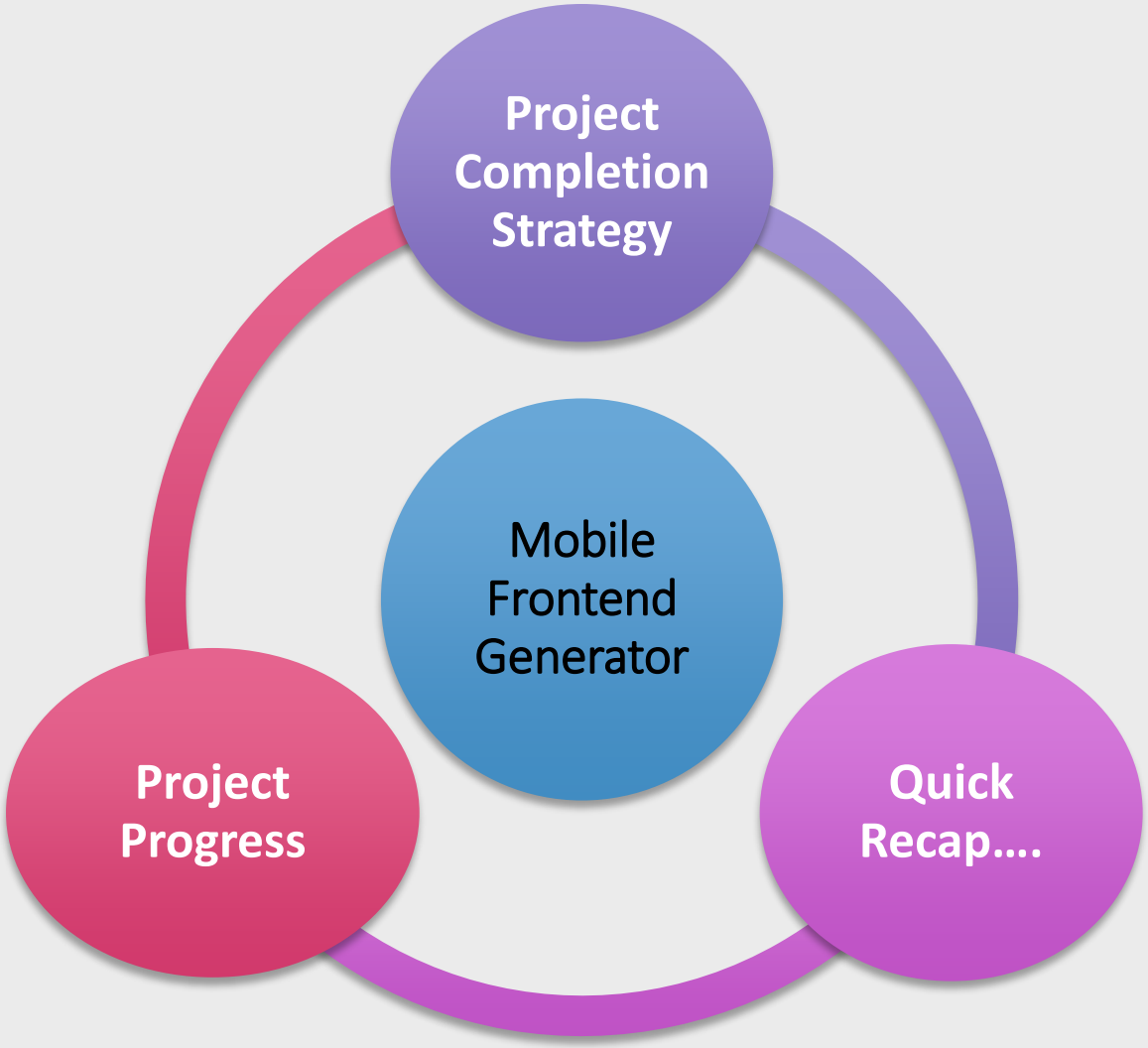
(SPECIALIZATION IN SOFTWARE ENGINEERING)

RESEARCH COMPONENT

FRONT-END CODE GENERATION FOR MOBILE APPLICATIONS USING CODE MAPPING



INTRODUCTION



QUICK RECAP.....

Research



- Introduce a mobile frontend generator module for ACG tool.

Research Question



- How to generate mobile frontend code for the given UI.

Objectives



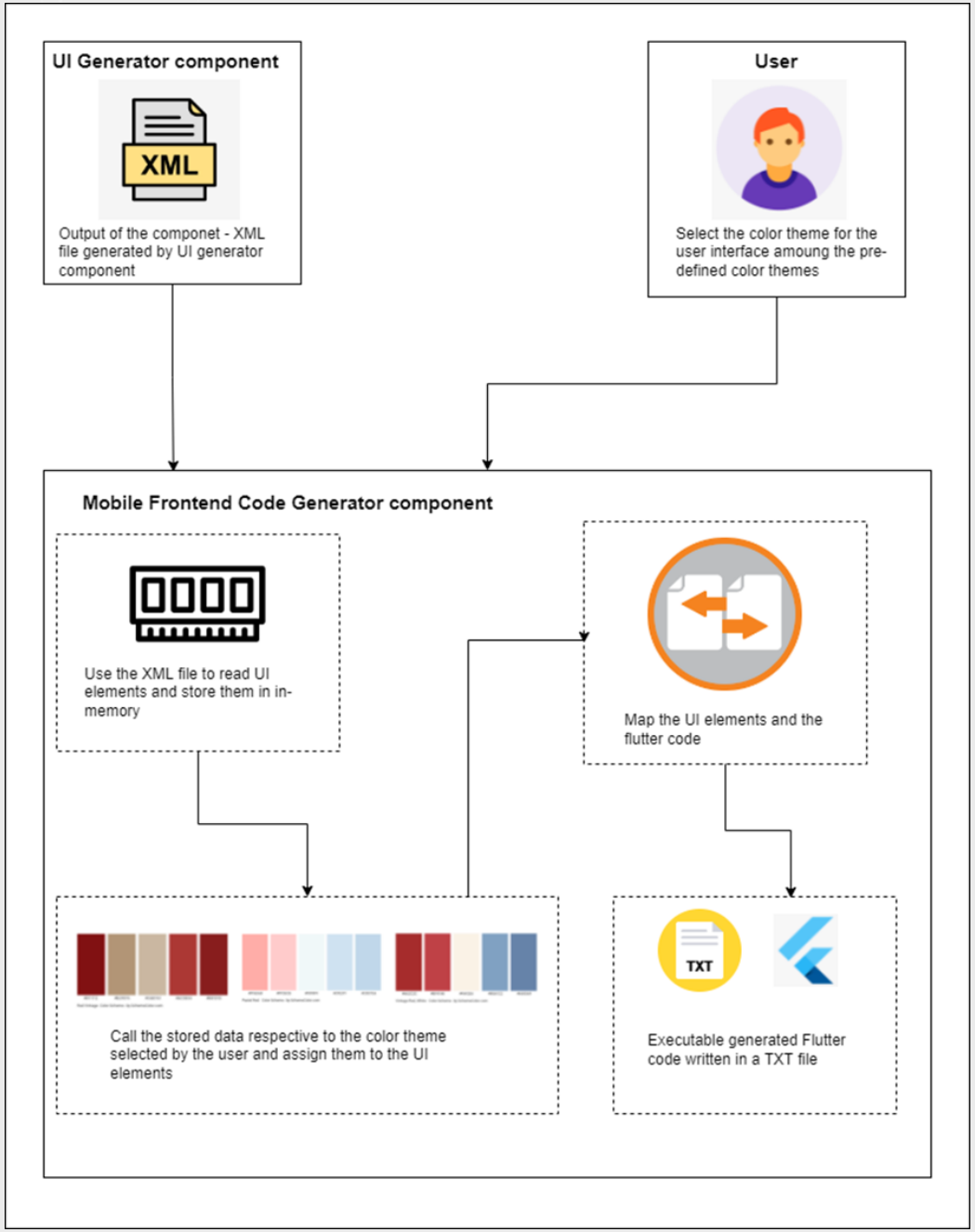
- Generate a simple and accurate automated mobile frontend code.

Research Gap

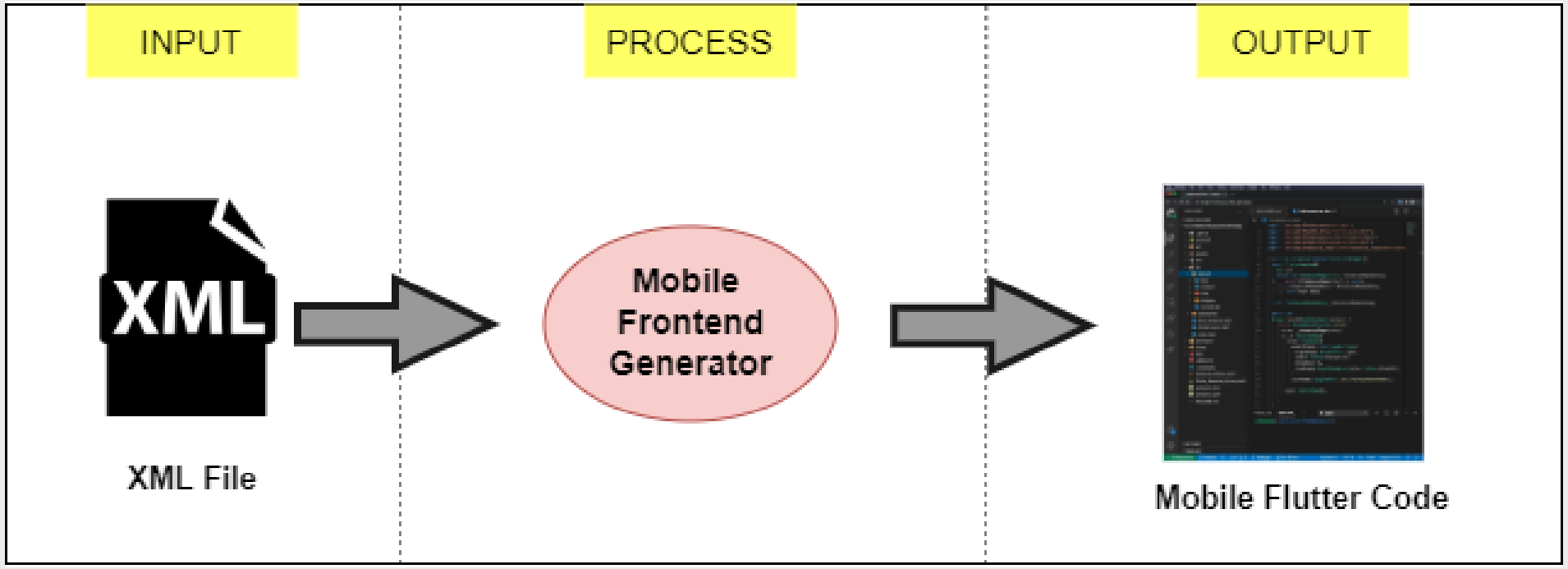


- Absence of flexible & scalable mobile frontend code generation process.

PROCESS


UI Generator component
XML
Output of the componet - XML file generated by UI generator component
User
Select the color theme for the user interface amoung the pre-defined color themes
Mobile Frontend Code Generator component
Use the XML file to read UI elements and store them in in-memory
Map the UI elements and the flutter code
Call the stored data respective to the color theme selected by the user and assign them to the UI elements
TXT
Executable generated Flutter code written in a TXT file

INPUT OUTPUT



PROGRESS COMPLETION

Read XML file

- Upload XML file
- Read UI elements

Add color theme

- Select the color theme from predefined themes
- Add the selected theme for the UI

Mobile frontend code generation

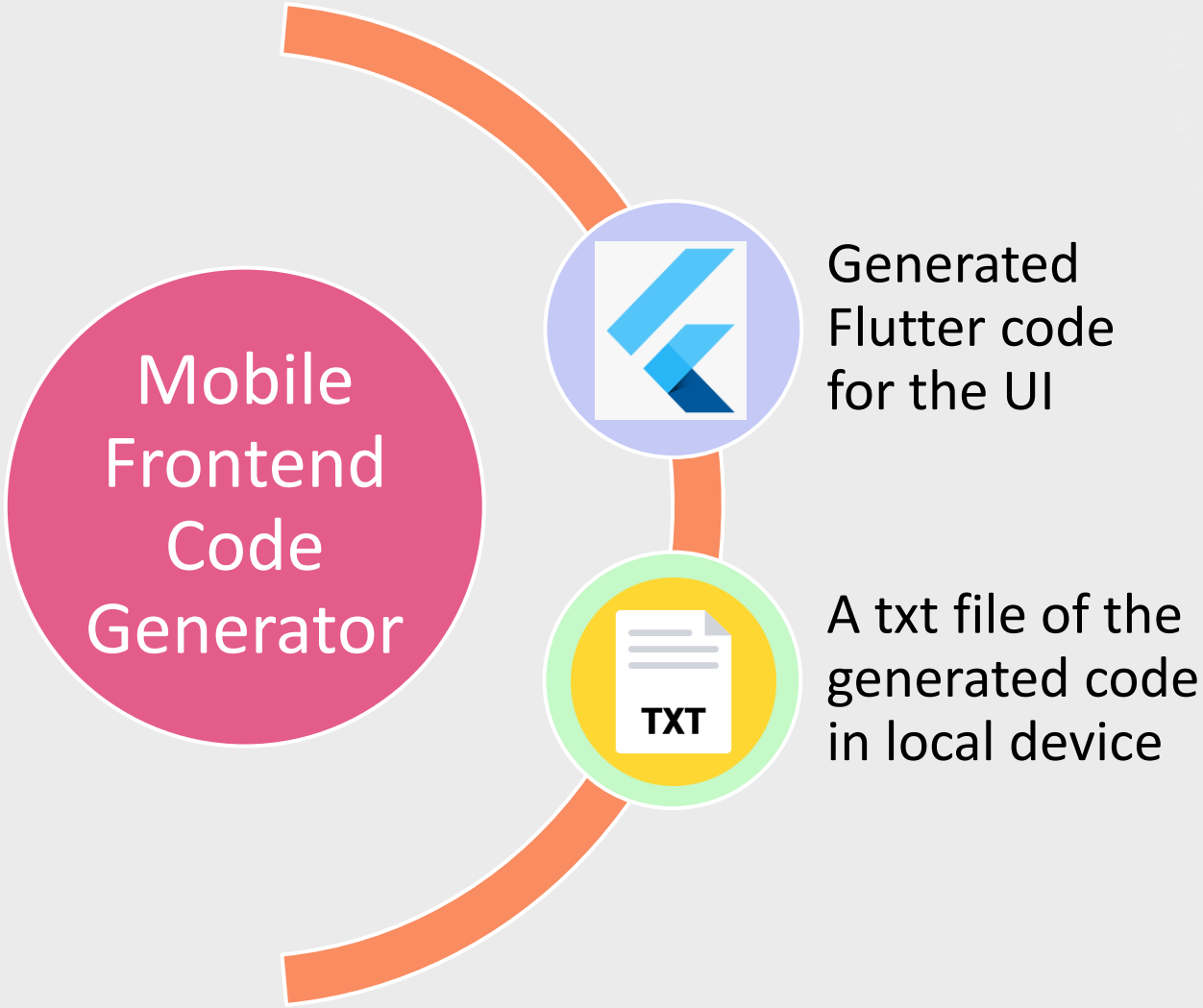
- Generate the mobile frontend code for the uploaded XML

Get the generated code

- Select the location where user needs to store the code
- View the same code from the application



OBSERVABLE OUTCOMES



Generated Flutter code for the UI

A txt file of the generated code in local device

SAMPLE GENERATED CODE - VIEW IN ANDROID STUDIO

main.dart

```dart
import 'package:flutter/material.dart';

void main() {
  runApp( MyApp());
}

class MyApp extends StatelessWidget {
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        backgroundColor: Color(0xFFffb997),
        appBar: AppBar(
          title: const Text('Welcome to My App'),
        ),  // AppBar
        body: MyCustomForm(),
      ),  // Scaffold
    );  // MaterialApp
  }}

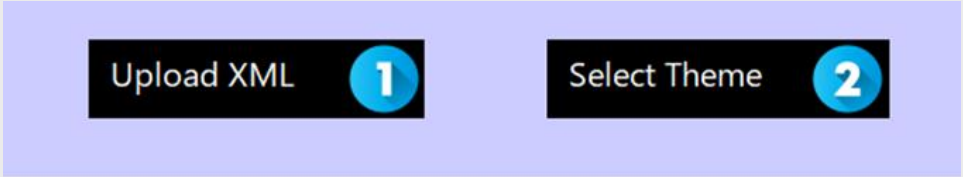
// Create a Form widget.
class MyCustomForm extends StatefulWidget{
  MyCustomFormState createState() {
    return MyCustomFormState();
  }
}

class MyCustomFormState extends State<MyCustomForm> {
```

main.dart

```dart
class MyCustomFormState extends State<MyCustomForm> {
  //a global key that uniquely identifies the Form widget and allows validation of the form
  final _formKey = GlobalKey<FormState>();
  Widget build(BuildContext context) {
    return Form(
      key: _formKey,
      child: Center( child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: <Widget>[
          const Text( 'Login Form',
              style: TextStyle(
                  color: Color(0xFF621940),
                  fontSize: 30,
                  fontWeight: FontWeight.bold)  // TextStyle
          ),  // Text
          const Divider(
              color: Color(0xFF0b032d)
          ),  // Divider
          const Text( 'Username',
              style: TextStyle(
                color: Color(0xFF000000),
                fontSize: 20,
              )  // TextStyle
          ),  // Text
          TextFormField(
            decoration: const InputDecoration(
```

SAMPLE GENERATED CODE – IN APPLICATION




```
import 'package:flutter/material.dart';

void main() {
        runApp( MyApp());
}

class MyApp extends StatelessWidget {
        Widget build(BuildContext context) {
        return MaterialApp(
                home: Scaffold(
                        backgroundColor: Color(0xFFc1e3ed),
                appBar: AppBar(
                        title: const Text('Welcome to My App'),
                ),
                body: MyCustomForm(),
                ),
        );
 }}

// Create a Form widget.
class MyCustomForm extends StatefulWidget{
        MyCustomFormState createState() {
                return MyCustomFormState();
        }
}

class MyCustomFormState extends State<MyCustomForm> {
        //a global key that uniquely identifies the Form widget and allows validation of the form
        final _formKey = GlobalKey<FormState>();
        Widget build(BuildContext context) {
         return Form(
                key: _formKey,
                child: Center( child: Column(
                        crossAxisAlignment: CrossAxisAlignment.start,
```

FUTURE WORK – GROUP

UI Generator

- ✓ Increase the number of UI components mapped in UI generator.

Web Frontend Generator

- ✓ Increasing the number of pre-define template styles and generating the web frontend-code according to other user-preferred technologies.

Mobile Frontend Generator

- ✓ Allow user to see the generated code in real time environment.

Backend & Database Generator

- ✓ Automatically bind the generated backend code with frontend code using AI.

COMMERCIALIZATION.

Silver — Standard

- Free.
- UI Generator and Backend code generator available.
- Maximum 4 projects.
- Code regeneration is disabled.

Gold — Community

- 23.50$ per year (Subscription based)
- UI Generator, Frontend code generator and backend code generator available,
- Maximum 15 projects within subscription period.
- Code regeneration and code refactor.

Platinum — Enterprise

- 50$ per year + 24x7 Customer support.
- All features are readily available.
- Unlimited projects.
- Offline support.



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THANK YOU..



Q & A.

DEMONSTRATION

